

A Riemannian Residual Learning Mechanism for SPD Network

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Abstract—The generalization of Euclidean network paradigm to the Riemannian manifolds has attracted much attention for offering useful geometric representations in processing manifold-valued data in recent years. However, the information degradation during data compression mapping hinders Riemannian networks from going deeper, and there are very few solutions specifically designed for this problem. Given the remarkable success of deep Residual learning in Euclidean networks, a novel Riemannian residual learning mechanism (RRLM) is proposed in the context of Symmetric Positive Definite (SPD) manifolds, enabling the characterization of deep spatiotemporal features while preserving the manifold properties. Based on RRLM, a stack of SPD manifold-constrained residual-like blocks is designed on the tail of the original SPDNet(backbone) for the sake of conducting deep Riemannian residual learning. For simplicity, we refer to the network architecture introduced above as Riemannian residual SPD network (ResSPDNet). The experimental results achieved on three types of visual classification tasks, *i.e.*, facial emotion recognition, drone recognition, and action recognition, demonstrate that our method can achieve improved accuracy with a deepened network structure.

Index Terms—Riemannian manifold, SPD matrix, Information degradation, Riemannian residual learning, Visual classification.

I. INTRODUCTION

Covariance matrices have achieved significant success in various image and video processing applications, such as facial emotion recognition [1]–[3], dynamic scene classification [4]–[6], and action recognition [7]–[10], owing to their capacity to globally capture temporal fluctuations of data sequences. However, traditional Euclidean computations are not suitable for covariance matrices, as the space spanned by them is adhered to a curved Riemannian manifold, also known as SPD manifold [11], [12]. Thanks to the development of Riemannian metric theory, the Euclidean tools can be generalized to learn the SPD features by utilizing the well-studied Riemannian metrics, such as Log-Euclidean metric (LEM) [11] and Affine-Invariant Riemannian metric (AIRM) [13]. Specifically, some SPD matrix learning methods provide a feasible path for solving the aforementioned issue by flattening the SPD manifold into an associated tangent space through tangent approximation [14]–[17]. Other algorithms exploit the Riemannian kernel functions [18]–[21] to embed the original manifold-valued features into a Reproducing Kernel Hilbert space (RKHS) that

satisfies the Euclidean geometry. However, the two approaches mentioned above are incapable of preserving the geometrical structure of the input data manifold well, as their representation learning process is basically carried out in the vector space. In order to address this problem, some geometry-aware metric learning algorithm are proposed [22]–[26]. They aim to represent the geometric information of the original data manifold more faithfully. However, the inherent linear feature transformation function of such approach is implemented on the nonlinear manifolds, resulting in the learned parameters being potentially suboptimal.

Compared with shallow learning methods, neural network-ing technique can endow feature representation learning with deep and nonlinear function. Encouraged by the merits of traditional deep learning methods, some recent works [1], [2], [5], [8], [10], [27] succeed in generalizing the construction paradigm of Euclidean networks to the domain of Riemannian manifolds to excavate more effective geometric semantic information for improved classification. Although the Riemannian neural networks such as SPDNet [1] have achieved better performance in many computer vision tasks, they are facing a degradation problem whereby the effectiveness of the learned features is weakened as the network depth increases [28].

To construct a deeper network capable of transforming the input SPD matrices into more informative representations with higher discriminability, a novel Riemannian residual learning mechanism (RRLM) is proposed in this article. Based on RRLM, we manage to extend the original SPD network to a deeper version that can effectively alleviate the problem of information degradation. Specifically, we first select SPDNet [1] as the backbone of the proposed network for the sake of extracting low-dimensional SPD representations with certain discriminability. At the end of the backbone network, an SPD feature learning module (SFLM) composed by multiple Riemannian residual-like blocks is designed (Fig. 1) to capture more critical structural information from the input data via deep residual learning. Besides, under the supervision of the reconstruction error term (RT) associated with the input and output SPD matrices of SFLM, the variability information contained in the input data will be well remained in the generated new feature manifolds. In this case, the loss function of the proposed model is capable of encoding and learning such information during end-to-end training adequately, being

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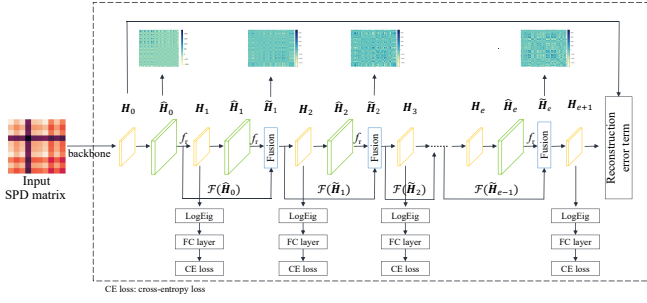


Fig. 1. The architecture of the proposed ResSPDNet.

conductive to alleviating the problem of model degradation preferably. It should be emphasized that each Riemannian residual block shown in Fig. 1 is constructed by shortcut connection (SC). However, different from the strategy that directly exploiting Element-wise Addition (EwA) to realize SC [28], the weighted Fréchet mean (wFM) [2] operation is adopted in this paper. Our motivations for doing this lie in the following four aspects: 1) Fréchet mean (Riemannian mean) has theoretical and practical merits in manifold data analysis [11]; 2) extending Fréchet mean to a weighted form could increase the discrimination of the generated features; 3) it can respect the Riemannian geometrical structure of SPD matrices faithfully; 4) it introduces very few parameters to be learned.

Considering the difference of statistical information contained in different hidden layers (H_e of the e^{th} ($e = 1 \rightarrow E$) building block), we connect a classification module consisting of the LogEig layer (will be introduced later), fully connected (FC) layer, and cross-entropy loss to H_e to enable the learning of a discriminative manifold-to-manifold embedding mapping by integrating different depth features. In such a case, a series of effective classifiers can be trained. Here, we adopt a maximum voting strategy for the final decision making. We conduct experiments on three different visual classification tasks, *i.e.*, video-based facial emotion recognition, skeleton-based action recognition, and drone recognition, respectively. The experimental results demonstrate that our ResSPDNet outperforms existing Riemannian deep neural networks in terms of classification accuracy.

II. RELATED WORKS

The geometry-aware SPD matrix learning methods [22]–[25] aim to preserve the Riemannian geometry of the input data manifold during feature transformation. However, their inherent linear embedding function makes it difficult to mine fine-grained geometric representations. This is particularly common in complicated visual scenarios. Motivated by the philosophy of convolutional neural networks (ConvNets) [29], [30], a Riemannian neural network (SPDNet [1]) is proposed to assign deep and nonlinear function to SPD matrix learning. Generally speaking, the construction paradigm of SPDNet is analogous to that of ConvNets for that each trainable block of SPDNet is comprised of a bilinear mapping layer and

an eigenvalue regularization layer for feature encoding and activation, respectively. On the basis of SPDNet, many new works [2], [3], [5], [10], [31], [32] are devoted to refurbishing the existing building blocks to adapt to different visual tasks.

Brooks et al. [2] reduces the training cost and alleviate the overfitting problem of SPDNet by introducing the manifold-valued parallel transport [33] into the network to achieve Riemannian batch normalization. However, the above networks are insufficient in mining local information embodied in the SPD matrix. To address this problem, Chakraborty et al. implement manifold-valued convolution by using wFM to simulate the weighted sum of CNN [31] and gain impressive results in medical image processing. To improve the classification performance of Riemannian deep learning methods, Wang et al. [5] design a lightweight cascaded neural network for SPD matrix nonlinear learning and classification, which achieves competitive training speed and accuracy, especially for the data limited scenario. However, the development of Riemannian deep learning is still in its infancy, and the degradation issue is one of the main bottlenecks affecting the deep representational capacity of the existing Riemannian networks. As a solution to this problem, Wang et al. [34] design a u-shaped SPDNet trained by a reconstruction error term, improving the effectiveness of the features generated by the hidden layer. Besides, this work enhances the information flow by connecting the encoder features with the decoder features, thereby the degradation problem can be alleviated.

These approaches demonstrate the potential of specialized network architectures to handle non-Euclidean data, and their success highlights the importance of developing such architectures for various applications.

III. PROPOSED METHOD

Extensive experimental results have demonstrated that endowing SPD matrix learning with deep and nonlinear function in the context of Riemannian manifolds is able to capture more effective feature representations with lower intra-class diversity and inter-class similarity. However, when simply increasing the network depth, the degradation of statistical information will appear, which results in the learned deep features cannot faithfully reflect the original data structure. In this case, the classification ability of the constructed deep Riemannian network will not be improved. In this paper, we propose a novel architecture to address this problem. Fig. 1 is an overview of our approach.

A. Preliminaries

This subsection aims to give a brief overview to the SPD manifold and the corresponding Log-Euclidean Metric (LEM) [11]. Besides, the covariance matrix modeling and the basic operation layers of SPD neural network are also introduced here.

SPD Manifold: A real-valued symmetric matrix $\mathbf{X}$ is called SPD if and only if for all non-zero vector $\mathbf{z} \in \mathbb{R}^d$, $\mathbf{z}^T \mathbf{X} \mathbf{z} > 0$. According to [11], we know that the space spanned by a set of

$d \times d$ SPD matrices is a specific Riemannian manifold, SPD manifold, represented by $\mathcal{S}_{++}^d$:

$$\mathcal{S}_{++}^d = \{\mathbf{X} \in \mathbb{R}^{d \times d} : \mathbf{X} = \mathbf{X}^T, \mathbf{z}^T \mathbf{X} \mathbf{z} > 0, \forall \mathbf{z} \in \mathbb{R}^d \setminus \{0_d\}\}. \quad (1)$$

It can be learned from [35] that $\mathcal{S}_{++}^d$ is a commutative Lie group and it admits a bi-invariant Riemannian metric. Therefore, the distance between two SPD matrices $\mathbf{X}_1, \mathbf{X}_2$ is given below:

$$d(\mathbf{X}_1, \mathbf{X}_2) = \|\log(\mathbf{X}_1) - \log(\mathbf{X}_2)\|_F, \quad (2)$$

where $\|\cdot\|_F$ is the Frobenius norm.

Data Modeling with Covariance Matrix: Denoting the i -th given data sequence (image set or video clip) as $\mathbf{F}_i = [\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n]$, where $\mathbf{f}_t \in \mathbb{R}^{d \times 1}$ signifies the t -th vectorized instance. The covariance representation of $\mathbf{F}_i$ can be computed by:

$$\mathbf{X} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{f}_i - \bar{\mathbf{f}})(\mathbf{f}_i - \bar{\mathbf{f}})^T, \quad (3)$$

where $\bar{\mathbf{f}} = \frac{1}{n} \sum_{t=1}^n \mathbf{f}_t$ is the mean of $\mathbf{F}_i$. To ensure that $\mathbf{X}$ satisfies the property of positive definite, we regularize it by $\mathbf{X} \leftarrow \mathbf{X} + \lambda \text{trace}(\mathbf{X}) \mathbf{I}$. Wherein, λ is a regularization coefficient configured as 0.001 in all the experiments, $\mathbf{I}$ is an identity matrix of size $d \times d$. In this way, a real SPD manifold element can be obtained, $\mathbf{X} \in \mathcal{S}_{++}^d$.

Basic layers of SPD network: Let $\mathbf{X}_{k-1} \in \mathcal{S}_{++}^d$ be the input SPD matrix of the k -th layer. The Riemannian operation layers of SPDNet [1] are introduced as follows:

BiMap Layer: This layer is designed to project the input SPD matrices into lower-dimensional feature representations by using a bilinear mapping function f_b , expressed as $\mathbf{X}_k = f_b^{(k)}(\mathbf{X}_{k-1}, \mathbf{W}_k) = \mathbf{W}_k^T \mathbf{X} \mathbf{W}_k$, where $\mathbf{W}_k$ is the to-be-learned transformation matrix with the constraint of column full-rank. In addition, $\mathbf{W}_k$ is also required to satisfy semi-orthogonality, such that a compact Stiefel manifold can be formed with the potential to render optimal solutions during optimization [1].

ReEig Layer: This layer is analogous to the classical ReLU layer, designed to endow SPDNet with a non-linear learning mechanism by using the following eigenvalue rectification function f_r , formulated as: $\mathbf{X}_k = f_r^{(k)}(\mathbf{X}_{k-1}) = \mathbf{U}_{k-1} \max(\epsilon \mathbf{I}, \boldsymbol{\Sigma}_{k-1}) \mathbf{U}_{k-1}^T$, where $\mathbf{X}_{k-1} = \mathbf{U}_{k-1} \boldsymbol{\Sigma}_{k-1} \mathbf{U}_{k-1}^T$ represents the eigenvalue decomposition, and ϵ is a small activation threshold.

LogEig Layer: This layer is designed to embed the learned SPD representations into a flat space using the following logarithmic mapping function f_l , so that the Euclidean learning methods can be applied. It is defined as: $\mathbf{X}_k = f_l^{(k)}(\mathbf{X}_{k-1}) = \mathbf{U}_k \log(\boldsymbol{\Sigma}_{k-1}) \mathbf{U}_k^T$, where $\mathbf{X}_{k-1} = \mathbf{U}_{k-1} \boldsymbol{\Sigma}_{k-1} \mathbf{U}_{k-1}^T$ denotes the eigenvalue decomposition, and $\log(\boldsymbol{\Sigma}_{k-1})$ signifies the eigenvalue logarithm of each diagonal element of $\boldsymbol{\Sigma}_{k-1}$.

B. RRLM-based Deep Riemannian Network

As shown in Fig. 1, the proposed ResSPDNet makes up of a cascade of Riemannian residual-like blocks (RRBs) to

facilitate the reconstruction of the remaining structural details, where the output feature matrices of the current RRB are used as the input data of the adjacent one. Owing to the introduced shortcut connections (SCs), the low-level features with rich structural information can be reused in the deep layers, enriching the information flow of ResSPDNet. Besides, under the supervision of the reconstruction error term, the augmented sequential connections are also beneficial for improving the feature learning capacity of the established deeper SPD network. Fig. 2 shows the detailed structure of the building block for Riemannian residual learning. From this figure, we can see that each RRB consists of the input (BiMap) layer, nonlinear activation (ReEig) layer, hidden (BiMap) layer for dimensionality reduction, upsampling (BiMap) layer for data reconstruction, and fusion layer for the implementation of SC. Considering that the BiMap-based upsampling operation can not maintain the positive definiteness of the generated data (rank deficient), another nonlinear activation layer is added to enable the final manifold-valued fusion operation.

Let $\mathcal{S} = [\mathbf{S}_1, \mathbf{S}_2, \dots, \mathbf{S}_N]$ be the original training set and $\mathbf{L} = [l_1, l_2, \dots, l_N] \in \mathbb{R}^{1 \times N}$ be the corresponding label vector. In this paper, the SPD manifold-valued training set is denoted as: $\mathcal{P} = [\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_N]$, where $\mathbf{P}_i$ is the i -th input SPD matrix of our network. For simplicity, we represent the feature matrix produced by the backbone as $\mathbf{Z}_i$. For the e -th Riemannian residual block, we use $\widetilde{\mathbf{H}}_{e-1}$ ($\widetilde{\mathbf{H}}_{e-1} = \hat{\mathbf{H}}_0$ when $e = 1$), $\mathbf{H}_e$, $\hat{\mathbf{H}}_e$, and $\widetilde{\mathbf{H}}_e$ to denote the input, output of the hidden layer, reconstruction of the input, and output of the fusion layer of the e -th ($e = 1 \rightarrow E$) RRB, respectively. In this context, $\mathbf{H}_e$ and $\hat{\mathbf{H}}_e$ can be computed by:

$$\mathbf{H}_e = f_b(\widetilde{\mathbf{H}}_{e-1}, \mathbf{W}_{e1}) = \mathbf{W}_{e1} \sigma(\widetilde{\mathbf{H}}_{e-1}) \mathbf{W}_{e1}^T, \quad (4)$$

$$\hat{\mathbf{H}}_e = f_b(\mathbf{H}_e, \mathbf{W}_{e2}) = \mathbf{W}_{e2}^T \mathbf{H}_e \mathbf{W}_{e2}, \quad (5)$$

where f_b denotes the bilinear mapping function, $\widetilde{\mathbf{H}}_{e-1}$ signifies the generated feature matrix of the fusion layer of the $(e-1)$ -th RRB, σ indicates the nonlinear activation operation, $\mathbf{W}_{e1} \in \mathbb{R}^{d_e \times d_{e-1}}$ ($d_e \leq d_{e-1}$) and $\mathbf{W}_{e2} \in \mathbb{R}^{d_e \times d_{e-1}}$ represent the transformation matrices of the e -th RRB to be learned. In this article, $\mathbf{W}_{e1}^T$ and $\mathbf{W}_{e2}^T$ are also endowed with the constraints of column full-rank and semi-orthogonality for better optimization and Riemannian geometry preservation.

According to the building block illustrated in Fig. 1, we can see that the SC enables the Riemannian residual learning to be adopted for every few stacked layers. Formally, in this paper, a building block is constructed as:

$$\widetilde{\mathbf{H}}_e = \mathcal{F}(\widetilde{\mathbf{H}}_{e-1}, \{\mathbf{W}_{e1}, \mathbf{W}_{e2}\}) \odot \widetilde{\mathbf{H}}_{e-1} \quad (6)$$

where $e = 2 \rightarrow E$ (when $e = 1$, $\widetilde{\mathbf{H}}_{e-1}$ is replaced by $\hat{\mathbf{H}}_0$), and $\mathcal{F}(\widetilde{\mathbf{H}}_{e-1})$ represents the Riemannian residual function to be approximated. For instance, $\mathcal{F} = \sigma(\mathbf{W}_{22}^T \mathbf{W}_{21} \widetilde{\mathbf{H}}_1 \mathbf{W}_{21}^T \mathbf{W}_{22})$ when e is set to 2, in which σ signifies the ReEig operation. Different from the traditional residual learning mechanism that works in the vector space, our method operates on SPD matrices with Riemannian geometry. Therefore, the fusion layer used to implement SC should be strictly defined in the

context of SPD manifolds. Taking the following two merits of Fréchet Mean (FM) [2], also known as Riemannian mean, into consideration: 1) it has shown strong theoretical and practical merits in manifold data analysis [13]; 2) it is faithful to the Riemannian properties of SPD matrix, we attempt to realize the manifold-valued SC between SPD matrices, symbolized as $\odot$, by extending the definition of FM to a weighted form.

Specifically, given a set of SPD matrices $\{M_i\}_{i=1}^{N_c}$ ($N_c = 2$ in this paper, where M_1 refers to $\widetilde{H}_{e-1}$, and M_2 refers to $\hat{H}_e$), the weighted Riemannian Mean can be formulated as:

$$G^* = \arg \min_{G \in \mathcal{S}_{++}^d} \sum_{i=1}^{N_c} w_i d^2(G, M_i), \quad (7)$$

where the weights need to meet the following conditions: 1) $w_i > 0$ for all $i \in \{1, \dots, N_c\}$; 2) $\sum_i w_i = 1$. At this time, the solution of Eq. (7) w.r.t LEM can be computed by:

$$G^* = \exp \left[\sum_{i=1}^{N_c} w_i \log(M_i) \right]. \quad (8)$$

In addition to exploiting wFM to realize SC, the Element-wise Addition (EwA) strategy used in ResNet is also theoretically feasible in this paper. The reason is that EwA can be used to process SPD matrices without destroying their Riemannian properties. Practically speaking, although the computational efficiency of wFM is lower than that of EwA, wFM is more faithful to the geometrical structure of SPD data points. As a consequence, the classification accuracy achieved by ResSPDNet-EwA will not be higher than that of ResSPDNet-wFM in theory. The following Table IV also verifies our speculation in the experimental scenario. Besides, there are very few parameters to learn in wFM.

C. Objective function

In short, this work aims to learn an effective manifold-to-manifold deep embedding mapping to transform the input SPD matrices into more informative and discriminative ones for improved classification. Considering the information degradation issue caused by increasing the depth of Riemannian network, a stack of Riemannian residual blocks is built on the tail of the backbone for the sake of fully mining the remaining structural details from the input stage-by-stage. In particular, wFM-based residual connections between SPD matrices can enhance the information flow of the network while preserving the manifold structure of the input data in the generated new feature space. As a result, more efficient and compact SPD matrices can be obtained to benefit classification. Moreover, the added reconstruction error term between the output feature matrices of the backbone and ResSPDNet enables the input data variations to be remained in the generated new feature manifolds, making the classification terms be more effective in encoding the geometric features with semantic information

of different perspectives. Hence, the objective function is given below:

$$\mathcal{L}(\theta_2, \phi, P) = \sum_{e=1}^E \sum_{i=1}^N \mathcal{L}_e(P_i, l_i) + \alpha \sum_{i=1}^N \mathcal{L}_2(H_0, H_{e+1}), \quad (9)$$

where $\theta_2 = \{\theta_1, W_{e1}, W_{e2}, T_e\}$ (T_e represents the projection matrix of the e -th FC layer), and α is the trade-off parameter, which will be discussed later.

The first term of Eq.(9) is the cross-entropy loss, defined as follows:

$$\mathcal{L}_e(P_i, l_i) = - \sum_{t=1}^c \mathcal{I}(l_i, t) \times \log \frac{e^{T_e^t V_e}}{\sum_{\tau} e^{T_e^\tau V_e}}, \quad (10)$$

where (P_i, l_i) denotes the input-target pairs, V_e signifies the vectorized H_e , T_e^t is the t -th row of the projection matrix $T_e \in \mathbb{R}^{c \times (d_e)^2}$, and $\mathcal{I}(l_i, t)$ is an indicator, where $\mathcal{I}(l_i, t) = 1$ if $l_i = t$, and 0 otherwise.

The second term is used to measure the reconstruction error between H_0 and H_{e+1} , given by:

$$\mathcal{L}_2(H_0, H_{e+1}) = \|H_0 - H_{e+1}\|_F^2, \quad (11)$$

where H_0 and H_{e+1} denote the output feature matrices of the backbone and entire network, respectively. To avoid matrix inversion, we use Euclidean distance to replace LEM for the computation of the reconstruction error term in this paper. We can learn from [28] that this substitution is feasible both experimentally and theoretically.

D. Riemannian Matrix Backpropagation

The loss of the k -th layer can be denoted by $L^{(k)} = \mathcal{L}_E \circ f^{(K)} \circ \dots \circ f^{(k)}$, where $\mathcal{L}_E$ is the cross-entropy loss of the E -th RRB. To optimize the proposed network, one of the essential steps is to update the weights in the BiMap layers on the compact Stiefel manifolds. By extending the stochastic gradient descent (SGD) algorithm to the context of Stiefel manifolds, the updating criteria for W_k [1] is given below:

$$\widetilde{\nabla} L_{W_k}^{(k)} = \nabla L_{W_k}^{(k)} - \nabla L_{W_k}^{(k)} W_k^T W_k, \quad (12)$$

$$W_k^{(t+1)} = \theta(W_k^{(t)} - \eta \widetilde{\nabla} L_{W_k}^{(k)}), \quad (13)$$

where $\widetilde{\nabla} L_{W_k}^{(k)}$ is the tangential component to the Stiefel manifold, computed by subtracting the normal component of the Euclidean gradient $\nabla L_{W_k}^{(k)}$, θ represents the retraction operation realized by the QR decomposition, W_k^t is the updated weight matrix at the t -th iteration, and η is the learning rate. For details about the optimization process, please kindly refer to [1].

Another essential step is to compute the gradients of SPD matrices in the wFM-based Fusion layer. According to Eq.(8), we define 3 auxiliary layers to enable the gradient computation, which are the LogEig layer, weighted-sum layer, and ExpEig layer. The LogEig layer is used to embed the manifold-valued data into a flat space, allowing for processing by the subsequent

Euclidean weighted-sum (WS) layer. The definition of the WS layer is given below:

$$\mathbf{X}_{k-1} = f_{ws}(\{w_i\}, \{\mathbf{X}_{k-2}^{(i)}\}) = \sum_{i=1}^{N_c} w_i \mathbf{X}_{k-2}^{(i)}, \quad (14)$$

where $\mathbf{X}_{k-2}^{(i)}$ represents the i -th input matrix of this layer ($i \in \{1, \dots, N_c\}$). In the backpropagation process, the partial derivatives of $L^{(k-1)}$ w.r.t w_i and $\mathbf{X}_{k-2}^{(i)}$ can be computed by:

$$\frac{\partial L^{(k-1)}}{\partial \mathbf{X}_{k-2}^{(i)}} = w_i \frac{\partial L^{(k)}}{\partial \mathbf{X}_{k-1}}, \quad \frac{\partial L^{(k-1)}}{\partial w_i} = \text{trace}(\mathbf{X}_{k-2}^{(i)} \frac{\partial L^{(k)}}{\partial \mathbf{X}_{k-1}}). \quad (15)$$

Then, the ExpEig layer is designed for the sake of mapping the generated data of the WS layer back onto the SPD manifold via the following equation:

$$\mathbf{X}_k = f_{exp}(\mathbf{X}_{k-1}) = \mathbf{U} \exp(\mathbf{\Sigma}) \mathbf{U}^T. \quad (16)$$

For the k -th ExpEig layer, it can be decomposed into two distinct parts. The first part is responsible for performing eigenvalue decomposition on the input matrix. In this case, a virtual intermediate layer k^* is introduced. Please kindly refer to [1], [36] for the detailed gradient computation in the k^* -th layer. In the second part, the inputs of the k -th layer are the eigenvalue matrix $\mathbf{\Sigma}$ and the corresponding eigenvector matrix $\mathbf{U}$, and the resulting matrix can be expressed as: $\mathbf{X}_k = \mathbf{U} \exp(\mathbf{\Sigma}) \mathbf{U}^T$. Since $d\mathbf{X}_k = 2(d\mathbf{U} \exp(\mathbf{\Sigma}) \mathbf{U}^T)_{sym} + (\mathbf{U} \exp(\mathbf{\Sigma}) d\mathbf{\Sigma} \mathbf{U}^T)_{sym}$, the following two partial derivatives can be obtained:

$$\frac{\partial L^{(k^*)}}{\partial \mathbf{U}} = 2 \left(\frac{\partial L^{(k+1)}}{\partial \mathbf{X}_k} \right)_{sym} \mathbf{U} \exp(\mathbf{\Sigma}), \quad (17)$$

$$\frac{\partial L^{(k^*)}}{\partial \mathbf{\Sigma}} = \exp(\mathbf{\Sigma}) \mathbf{U}^T \left(\frac{\partial L^{(k+1)}}{\partial \mathbf{X}_k} \right)_{sym} \mathbf{U}. \quad (18)$$

For the detailed derivation process of Riemannian matrix backward pass in the BiMap, ReEig, and LogEig layers, please refer to [1]. According to the previous introduction, we summarize the main implementation details of the proposed ResSPDNet into **Algorithm 1**.

E. Motivation for Designing the Riemannian Residual Block

Given that the weight matrices $\mathbf{W}_k$ are semi-orthogonal ($\mathbf{W}_k^T \mathbf{W}_k = \mathbf{I}$), it can be inspired from the traditional autoencoder paradigm to design a Riemannian autoencoder consisting of successive layers of SPD matrix downsampling and upsampling that theoretically approaches an identity map (IM). For simplicity, we regard $\tilde{\mathbf{H}}_2 = \mathbf{W}_2^T \mathbf{W}_1 \tilde{\mathbf{H}}_1 \mathbf{W}_1^T \mathbf{W}_2$ as the output SPD matrix after one downsampling and upsampling layer, shown in Fig. 2. In theory, under the supervision of the reconstruction error term, $\mathbf{W}_1$ and $\mathbf{W}_2$ will tend to be closer to each other. Besides, the operation mechanism of the ReEig layer makes it bring about minor changes to the eigenvalue space of each input SPD matrix. As a consequence, it can be inferred that $\|\tilde{\mathbf{H}}_2\|_F \rightarrow \|\tilde{\mathbf{H}}_1\|_F$. In this case, it can be expected that the constructed deeper model will not yield lower accuracy than that of the shallow backbone. The experimental results presented in the following Fig. 4 also

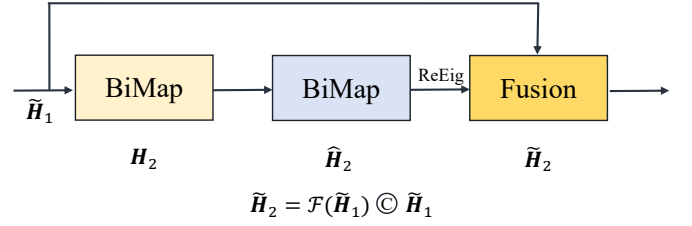


Fig. 2. Riemannian residual learning block (RRB).

verify this. Furthermore, the introduced shortcut connections make the current RRB learning phase can access the features of the previous stage directly, facilitating the reconstruction of the remaining residuals. Since the weight space is a compact stiefel manifold, it is impossible for the solver to optimize each weight matrix toward zero. In other words, the optimal Riemannian residual function to be learned in this paper will not be a zero mapping. The following Fig. 5 demonstrates that the Riemannian residual mapping actually performs a continuous incremental learning on the structural information of the input data, which is essentially different from the hypothesis verified in ResNet, *i.e.*, the to-be-learned residual function is close to a zero mapping.

IV. EXPERIMENTS

In this section, we evaluate the classification performance of the proposed ResSPDNet on three different visual classification tasks, including drone recognition on the Radar dataset [37], facial emotion recognition on the AFEW dataset [38], and action recognition on the HDM05 dataset [2].

A. Implementation

In this article, we employ 4 layers to construct the backbone network: $\mathbf{P}_i \rightarrow f_b^{(1)} \rightarrow f_r^{(2)} \rightarrow f_b^{(3)} \rightarrow f_r^{(4)}$, where f_b and f_r denote the BiMap layer and ReEig layer, respectively. The proposed ResSPDNet is comprised of E RRBs, each RRB makes up of: $\tilde{\mathbf{H}}_{e-1} \rightarrow f_b$ (hidden) $\rightarrow f_b$ (reconstruction) $\rightarrow f_r \rightarrow f_{fusion}$. Besides, the hidden layer of each RRB also connects to a classification module, consisting of 3 layers: f_l (LogEig) $\rightarrow f_{fc}$ (FC) $\rightarrow f_{CE}$ (Cross-Entropy loss). In our experiments, the learning rate η , batch size B , threshold ϵ of the ReEig layer are set to 0.01, 30, and $1e-4$ for all datasets, respectively. To enable the manifold-valued optimization, the weights of the BiMap layers are initialized as random semi-orthogonal matrices.

B. Dataset Description and Settings

Radar Dataset: A radar signal is the result of an emitted wave reflecting off the target. For this dataset, each signal is split into windows of length 20, rendering a 20×20 nonsingular covariance matrix for each window. The synthesized dataset consists of 500 data points per class. On this dataset, we set the sizes of the transformation matrices w.r.t the backbone to 20×16 , and those of the e -th residual block are configured as 16×8 and 8×16 , respectively.

Algorithm 1 ResSPDNet

Input: Training set $\mathcal{S}$, training epochs $\mathcal{T}$, learning rate η , batch size B , number of B of a given dataset $\mathfrak{b}$, threshold ϵ , number of layers K , regularization parameter λ and trade-off parameter α .

Initialization:

for $i \leftarrow 1$ **to** N **do**:

Computing $\mathbf{X}_i$ from $\mathbf{F}_i$ via $\mathbf{X} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{f}_i - \bar{\mathbf{f}})(\mathbf{f}_i - \bar{\mathbf{f}})^T$,

Using $\mathbf{X} = \mathbf{X} + \lambda \text{trace}(\mathbf{X})\mathbf{I}$ to ensure that $\mathbf{X}_i$ is SPD.

end

for $k \leftarrow 1$ **to** K **do**:

The SVD operation is applied to initialize $\mathbf{W}_k$ of the BiMap layer.

end

Training:

for $r \leftarrow 1$ **to** $\mathcal{T}$ **do**:

Randomly and equally divide the new training set $\mathcal{X}$ into $\mathfrak{b}$ batches:

$\mathcal{X} = \text{random}[\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_{\mathfrak{b}}]$.

for $h \leftarrow 1$ **to** $\mathfrak{b}$ **do**:

for $k \leftarrow 1$ **to** K **do**:

Forward Pass:

case 'BiMap layer'

$\mathbf{X}_k = f_b^{(k)}(\mathbf{W}_k, \mathbf{X}_{k-1}) = \mathbf{W}_k^T \mathbf{X}_{k-1} \mathbf{W}_k$.

case 'ReEig layer'

$\mathbf{X}_k = f_r^{(k)}(\mathbf{X}_{k-1}) = \mathbf{U} \max(\epsilon \mathbf{I}, \Sigma) \mathbf{U}^T$.

case 'LogEig layer'

$\mathbf{X}_k = f_l^{(k)}(\mathbf{X}_{k-1}) = \mathbf{U} \log(\Sigma) \mathbf{U}^T$.

case 'Shortcut connection'

$\mathbf{G}_k^* = \arg \min_{\mathbf{G}_k \in \mathcal{S}_{++}^d} \sum_{i=1}^N w_i d^2(\mathbf{G}_k, \mathbf{M}_i)$.

case 'Cross-Entropy loss'

$\mathcal{L}_e = - \sum_{i=1}^N \sum_{t=1}^c \mathcal{I}(l_i, t) \times \log(P(t|\mathbf{X}_i))$.

case 'Reconstruction error term'

$\mathcal{L}_2(\mathbf{H}_0, \hat{\mathbf{H}}_{e+1}) = \|\mathbf{H}_0 - \hat{\mathbf{H}}_{e+1}\|_{\mathbb{F}}^2$.

Backward Pass:

case 'Shortcut connection'

Computing $\frac{\partial \mathcal{L}_2}{\partial \mathbf{X}_{k-1}}^{(k)}$ using Eq. (15), Eqs. (17-18).

case 'reconstruction error term'

$\frac{\partial \mathcal{L}_2}{\partial \mathbf{H}_0} = 2(\mathbf{H}_0 - \hat{\mathbf{H}}_{e+1}), \frac{\partial \mathcal{L}_2}{\partial \hat{\mathbf{H}}_{e+1}} = -2(\mathbf{H}_0 - \hat{\mathbf{H}}_{e+1})$.

end

end

end

Output: $\mathcal{W} = \{\mathbf{W}_K, \mathbf{W}_{K-1}, \dots, \mathbf{W}_2, \mathbf{W}_1\}$.

HDM05 Dataset: This dataset contains 2083 SPD matrices belonging to 130 different classes. Following the previous works [1], [11], [39], we transform each frame into a 93-dimensional vector using the provided 3D coordinates of 31 hand joints. In the experiments, we set the sizes of the transformation matrices of the backbone and each residual

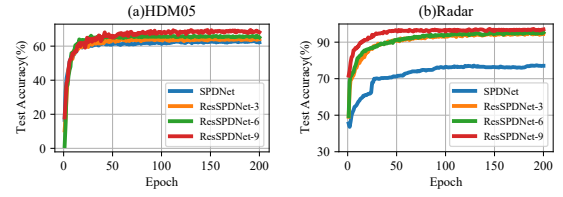


Fig. 3. The accuracy of SPDNet/ResSPDNet-3/ResSPDNet-6/ResSPDNet-9 versus the number of training epochs on the HDM05 and Radar datasets.

TABLE I
RESULTS ON THE RADAR DATASET.

Metric	ResSPDNet-3	ResSPDNet-6	ResSPDNet-9
Acc.(%)	94.30	96.20	97.40
s/epoch	12.10	20.84	34.98

TABLE II
RESULTS ON THE HDM05 DATASET.

Metric	ResSPDNet-3	ResSPDNet-6	ResSPDNet-9
Acc. (%)	64.58	67.24	69.17
s/epoch	49.02	95.30	131.45

TABLE III
RESULTS ON THE AFEW DATASET.

Metric	ResSPDNet-3	ResSPDNet-6	ResSPDNet-9
Acc.(%)	34.56	35.87	36.44
s/epoch	118.95	212.34	340.25

block to 93×63 and $\{63 \times 30, 30 \times 63\}$, respectively.

AFEW Dataset: This dataset consists of videos depicting 7 classes of facial expressions. For evaluations, we follow the protocol of [1], [11], [39] to divide it into training, validation and testing sets, firstly. Then, the training set is segmented into 1747 clips for data augmentation. Finally, we normalize each facial frame to an image of 20×20 , resulting in that a covariance matrix of 400×400 can be computed for the representation of each video clip. On this dataset, the sizes of the connection weights are set to 400×200 and 200×100 for the backbone and those for the e -th residual block are configured as 100×50 and 50×100 .

C. Ablation Studies

In this section, we conduct experiments to evaluate the reliability and effectiveness of the proposed method in visual classification.

Ablation study for ResSPDNet: We first carry out experiments on the three used datasets to study the impact of the network depth on the classification performance of our method. From Table I, we can observe that the classification accuracy of ResSPDNet-6 is higher than that of ResSPDNet-3, which validates that the proposed Riemannian residual learning mechanism (RRLM) can help alleviate the degradation problem and achieve accuracy gains. We can draw the same experimental findings from Table II and Table III. Secondly, we

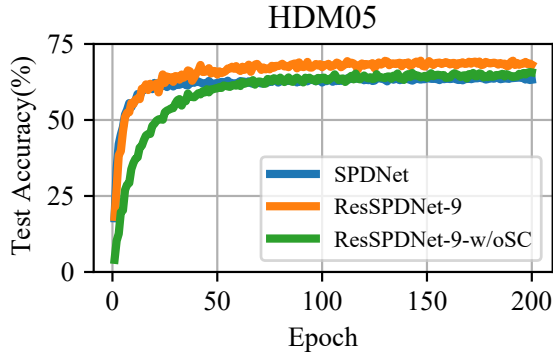


Fig. 4. The accuracy of SPDNet/ResSPDNet-9/ResSPDNet-9-w/oSC versus the number of training epochs on the HDM05 dataset, where w/oSC means shortcut connections are not used.

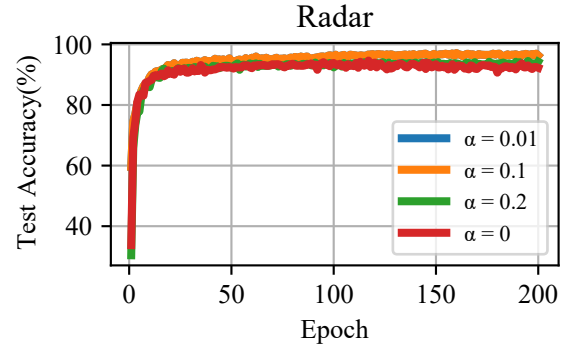


Fig. 6. Accuracy comparison on the Radar dataset under different values of α .

TABLE IV
COMPARISON UNDER DIFFERENT DATA FUSION METHODS ON THE RADAR DATASET.

Methods	Element-wise Addition (EwA)	wFM
Acc. (%)	93.06	94.30

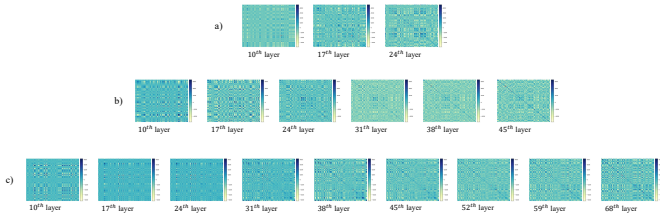


Fig. 5. The feature maps from different fusion layers of ResSPDNet-3/ResSPDNet-6/ResSPDNet-9 on the HDM05 dataset are visualized in (a), (b) and (c), respectively.

also explore ResSPDNet-9 by simply increasing the number of the constructed Riemannian residual blocks. It can be found that the classification ability of ResSPDNet-9 has been further improved, compared to ResSPDNet-3 and ResSPDNet-6 on the Radar, HDM05, and AFEW datasets. However, the training time (s/epoch) listed in the three tables indicates that the increased depth of network has a detrimental effectiveness on its computational efficiency.

Visualization: In order to intuitively verify the effectiveness of the proposed method in enhancing the information flow and mitigating the problem of information degradation, we visualize the SPD feature maps learned by ResSPDNet at different fusion layers on the HDM05 dataset. As can be observed from Fig. 5 that the high-level feature matrices have a more concentrated energy, especially the features on the main diagonal. This is indicated by the above phenomena that the statistical features embodied in the original visual scene can be effectively captured by the proposed network, thus being beneficial for visual classification.

The visualized feature maps also certify that the to-be-learned Riemannian residual mapping $\mathcal{F}$ does not approach the zero mapping, which is inconsistent with the hypothesis presented

by ResNet [29]. The fundamental reason is that it is impossible for the Riemannian solver to drive the weight matrices lying in the compact Stiefel manifold toward a zero matrix. We can learn from Fig. 5 that the established Riemannian residual function actually plays a role in incremental learning of the remaining structural information of the input data, verifying the previous theoretical analysis experimentally.

Ablation study for the shortcut connections: We take the HDM05 dataset as an example to investigate the impact of the shortcut connections (SCs) on the classification ability of the proposed approach. From Fig. 4, it is evident that ResSPDNet-9-w/oSC obtained by removing the SCs from the established Riemannian residual blocks has a good convergence behavior, but its classification accuracy is lower than that of the original model. This suggests that the SC-based feature reusing between different Riemannian residual blocks is capable of strengthening the information flow of the proposed network, thereby helping to alleviate the problem of information degradation caused by increasing the network depth. As mentioned in Section III-B, in addition to wFM, the Element-wise Addition (EwA) operation, which is often used in Euclidean networks, can also be exploited to implement SC. Therefore, we make experiments to evaluate the merits of wFM over EwA, choosing the Radar dataset as an example. The experimental results listed in Table IV show that the accuracy of the proposed ResSPDNet embedded with wFM is 1.24% higher than that of using EwA, further certifying that wFM is a more respectful way in maintaining the Riemannian geometry of the input data manifold.

Ablation study for reconstruction error term: As mentioned before, the reconstruction error term (RT) is able to promote the embedding mechanism of the constructed SPD feature learning module at the end of the backbone to approach an identity mapping, which is beneficial for learning more informative features. To demonstrate this in the experimental scenario, we conduct experiments on the Radar dataset as an example to study the impact of RT on the classification accuracy of the proposed network.

From Fig. 6, we can find that when the trade-off parameter α is assigned to a larger value, the network will pay more attention to learn deep reconstruction features, which

TABLE V
COMPARISON ON THE HDM05 DATASET, WHERE 'CM' DENOTES THE CLASSIFICATION MODULES.

Methods	Accuracy (%)
1st CM	62.14
2nd CM	62.89
3rd CM	64.21
1st & 2nd CMs	63.23
1st & 3rd CMs	64.34
2nd & 3rd CMs	64.37
1st, 2nd and 3rd CMs	64.58

is detrimental to the classification task. The relatively lower accuracy achieved by the proposed method at $\alpha = 0.2$ verifies this. According to Fig. 6, it is also evident that the curve corresponding to $\alpha = 0$ is located below other curves and our model is less sensitive to this parameter, rendering that RT helps to fine-tune the classification ability of the suggested network.

The aforementioned experimental observations certify that RT can provide some complementary information related to the data structure for the classification module so that it can make better decisions. In this ablation study, our rule for choosing the value of α is to make the reconstruction error term at least one order of magnitude lower than that of the classification term. Based on this criterion, the softmax classifier can better integrate the gradient information w.r.t RT to learn a more reasonable hypersphere for different classes.

Ablation study for the classification module: In this subsection, we conduct experiments on the HDM05 dataset to investigate the impact of the number of classification modules (CMs) used in the training stage on the accuracy of the proposed ResSPDNet. Considering that ResSPDNet-3 can induce 7 different combinations of CMs, we take it as an example in this study. According to the results given in Table V, we can have the following two observations: 1) the more the number of classifiers, the higher the classification accuracy; 2) the features produced by the third classifier are more effective than the others. This implies that the used classifiers are complementary to each other in providing useful supervised information for describing class distribution.

D. Comparison of with the state-of-art methods

In this subsection, we compare the proposed ResSPDNet with some representative methods in terms of classification accuracy on the three datasets involved in this article. The experimental results of the different methods achieved on the Radar and HDM05 datasets are listed in Table VI. For a fair comparison, we run the source codes of SPDNet and SPDNetBN by following the original recommendations to adjust the parameters on these two datasets, and report their best results. From Table VI, it is evident that the classification score of ResSPDNet-3 is respectively 21.14% and 6.44% higher than that of SPDNet and SPDNetBN on the Radar dataset. Besides, as the network depth increases, a continuous accuracy gains can be obtained on the Radar dataset. According to Table VI,

TABLE VI
ACCURACY (%) COMPARISON ON THE RADAR AND HDM05 DATASETS.

Methods	Source	Radar	HDM05
SPDNet [1]	AAAI'17	73.16	61.84
SPDNetBN [2]	NeurIPS'19	87.86	65.49
ResSPDNet-3		94.30	64.58
ResSPDNet-6		96.20	67.24
ResSPDNet-9		97.40	69.17

TABLE VII
ACCURACY (%) COMPARISON ON THE AFEW DATASET.

Methods	Source	Accuracy(%)
ManifoldNet [31]	TPAMI'22	23.98
DARTS [40]	ICLR'19	25.87
GEMKML [18]	TMM'21	35.71
SymNet [5]	TNNLS'22	32.70
GrNet [41]	AAAI'18	34.23
SPDNet [1]	AAAI'17	34.23
SPDNetBN [2]	NeurIPS'19	36.12
ResSPDNet-3		34.56
ResSPDNet-6		35.87
ResSPDNet-9		36.44

we can also find that the classification ability of ResSPDNet-6/9 is superior to that of SPDNet and SPDNetBN on the HDM05 dataset, again demonstrating the effectiveness of our approach in tackling the degradation problem. However, the classification accuracy of SPDNetBN is 0.91% higher than that of ResSPDNet-3. Since the Riemannian batch normalization operation conveys the ideology of multi-stage metric learning, it is able to encode and learn the feature distributions better than ResSPDNet-3.

Here, we also choose several state-of-the-art methods for better comparison on the AFEW dataset. Note that the experimental results of all the comparative methods tabulated in Table VII are obtained by running their official implementations with default settings. Besides, we implement DARTS in the domain of SPD matrix logarithms for a fair comparison. According to Table VII, we can observe that the classification scores of ResSPDNet-3/6 are higher than SPDNet and most of the competitors. This further supports our theoretical analysis that the designed Riemannian residual blocks can perform incremental learning on the input data structure. From Table VII, we can also find that ResSPDNet-9 achieves the highest accuracy on the AFEW dataset. All in all, the aforementioned experimental observations show that the proposed Riemannian residual learning mechanism can help training an effective deeper network.

V. CONCLUSION

In this paper, we propose an effective method to solve the degradation problem of SPD neural networks. This is achieved by designing a novel cascading network architecture consisting of a stack of Riemannian residual blocks added to the tail of the backbone SPD network. The experimental results suggest

that the proposed Riemannian residual learning mechanism is a resultful strategy against model degradation and is able to achieve accuracy gains with increasing the network depth. For future work, it would be interesting to generalize the paradigm of dense connection to the context of SPD networks using the wFM-based feature concatenation. Theoretically, the enriched information flow will not only be profitable to the training of deep networks, but also be beneficial to yield more informative deeper features.

VI. ACKNOWLEDGMENT

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